Assessing head and neck

Weber: Chapter 15

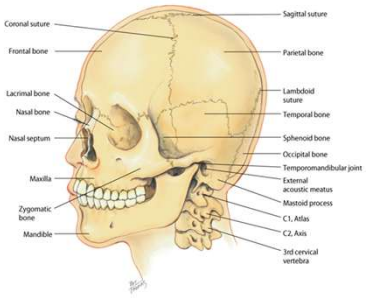
NURS3120: Health Assessment and Promotion

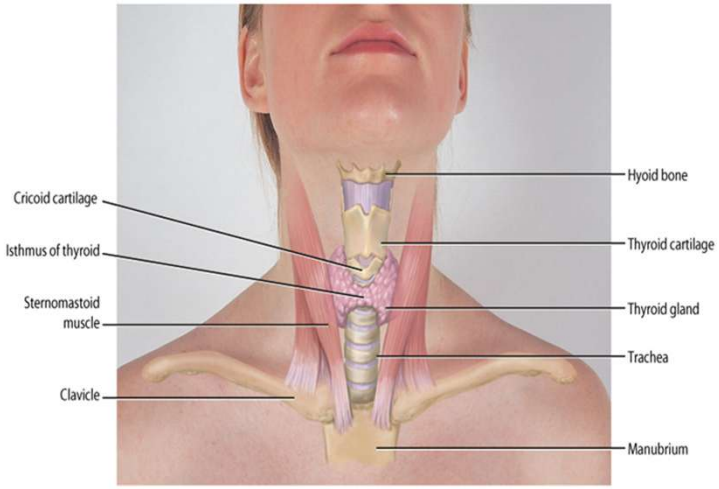
Objectives

1. Describe the structures and functions of the head and neck
2. Use the correct technique to perform a physical assessment of the head and neck
3. Differentiate normal vs abnormal findings related to the head and neck
4. Analyze the client's interview and physical assessment head and neck data to formulate valid clinical judgments

2

Anatomy of head & neck





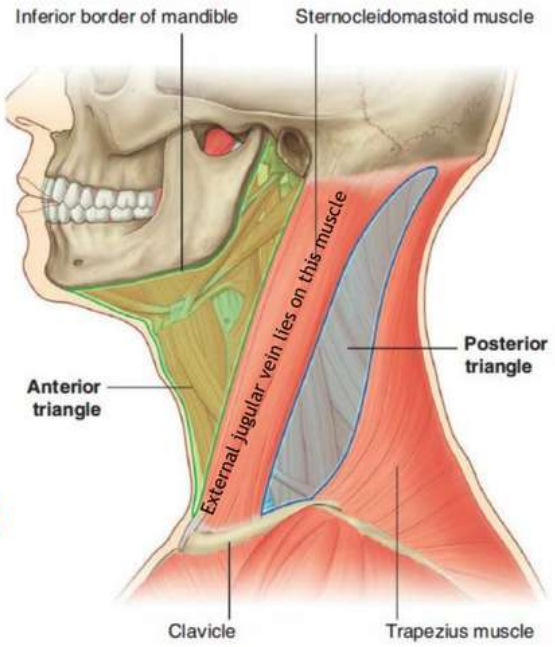
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The head: the skull which is divided into two subsections: the cranium and the face. The cranium houses and protects the brain and major sensory organs. It consists of eight bones: Frontal, parietal, temporal, occipital, ethmoid, sphenoid. There are 14 facial bones: 2 maxilla, 2 zygomatic, 2 inferior conchae, 2 nasal, 2 lacrimal, 2 palatine, 1 vomer, 1 mandible jaw.

The structure of the neck is composed of muscles, ligaments, and the cervical vertebrae. Contained within the neck are the hyoid bone, several major blood vessels, the larynx, trachea, and the thyroid gland, which is in the anterior triangle of the neck.

The neck muscles



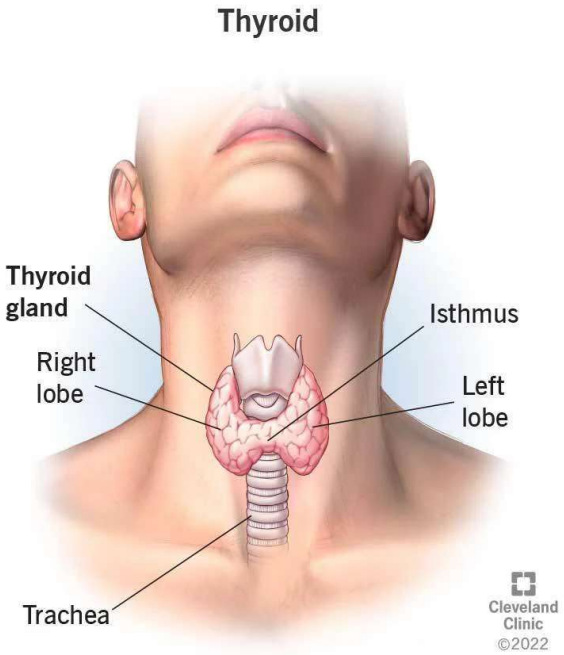


4

The sternomastoid (sternocleidomastoid) and trapezius muscles are two of the paired muscles that allow movement and provide support to the head and neck. These two muscles also form two triangles that provide important landmarks for assessment. The anterior triangle is located under the mandible, anterior to the sternomastoid muscle. The posterior triangle is located between the trapezius and sternomastoid muscle

Thyroid gland

- Largest endocrine gland in the body
- Produces thyroid hormones that increase the metabolic rate of most body cells
- 2 lobes that curve posteriorly on both sides of the trachea & esophagus
- Mostly covered by sternomastoid muscles
- 2 lobes connected by the isthmus



5

Lies in the lower half of the neck, just anterior to the trachea

Lymph nodes in the neck

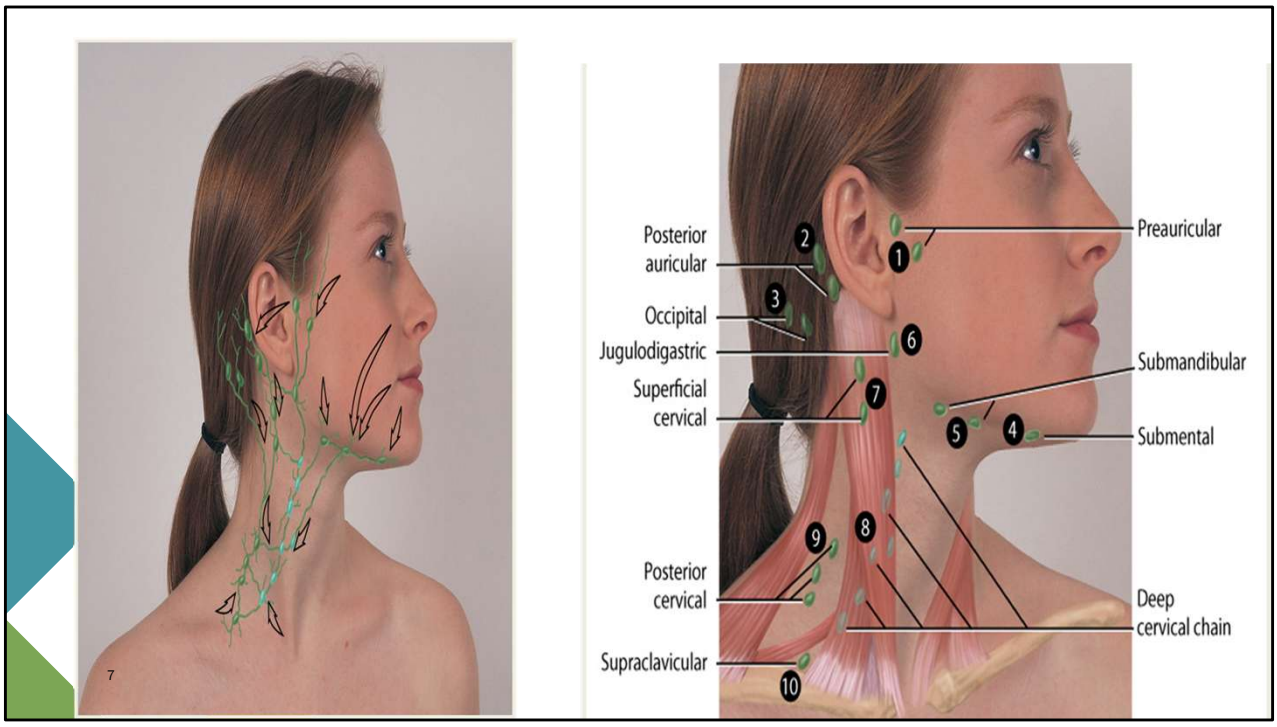
Lymphatic system

- An extensive vessel system & a major part of the immune system, which detects & eliminates foreign substances from the body
- Vessels allow for flow of clear, watery fluid from tissue spaces into circulation
- Greatest supply is in the head & neck`

- Nodes are small, oval clusters of lymphatic tissue that filter lymph & engulf pathogens, preventing potentially harmful substances from entering the circulation
- You should be familiar with direction of drainage patterns of lymph nodes
- Assess: Head, neck, arms, axillae, & inguinal area

6

There are several lymph nodes located in the head and neck. Lymph nodes filter lymph, a clear substance composed mostly of excess tissue fluid after the lymphatic vessels collect it but before it returns to the vascular system. Filtering removes bacteria and tumor cells from lymph. In addition, the lymph nodes produce lymphocytes and antibodies as a defense against invasion by foreign substances. The size and shape vary, but most are less than 1cm long and are buried deep in the connective tissue. Normally lymph nodes are either not palpable or they may feel like very small beads. So if a lymph node is overwhelmed by microorganisms, as happens with an infection, they swell and become painful. (Lymphadenopathy). If cancer metastasizes to the lymph nodes, they may enlarge but are not painful. Normal findings of lymph nodes: soft, moveable, and non-tender OR nonpalpable.



The picture on the left is the drainage system of the lymph nodes which is marked by the black arrows and the green is the location of each lymph node and the picture on the right is the lymph nodes and their location.

Collecting health history: Subjective data

- Do you get any headaches?
- Head injury?
- Dizziness?
- Neck pain or limitation of neck motion?
- Lumps or swelling?
- Protective head gear with activity?

- Head or neck surgery?
- Do you smoke or chew tobacco?
- Any fatigue?
- Weight gain?
- Bradycardia?
- Weakness?
- Palpitations?
- Excessive sweating?



8

If they have pain, what are you going to ask or what mnemonic? COLDSPA
Neck pain may accompany muscular problems or cervical spinal cord problems, or in the older adults that have arthritis or osteoporosis may experience neck pain and decreased range of motion. Limitation of neck movement could also be a vertebral joint dysfunction.
What would lumps or swelling lead you to think about? Cancer? Or a goiter? Enlarged thyroid gland which could affect a client's ability to swallow if big enough.
Client should wear protective gear when participating in recreational activities to protect the head and neck from injury.
Sudden trouble seeing or visual disturbances in one or both eyes or trouble walking, dizziness, or loss of balance or coordination may be a sign of an impending stroke
Head injury? Chronic pain or limited movement because of a previous surgery or trauma.
Smoking increases the risk of head and neck cancer

Physical assessment

Inspect the head & face

- Note size and shape of head
- Note symmetry of the face

Palpate the head & face

- Palpate head for lesions or lumps
- Temporal region/Temporal artery
- Temporomandibular joint

9

Inspect the head for size, shape and configuration. Usually the head is symmetric, round, erect and in midline and appropriately related to body size. Note any lesions. ABNORMAL: Acorn-shaped, enlarged skull bones are seen in Paget disease. Inspect for involuntary movements of the head. Lesions or lumps on the head may indicate recent trauma or sign of cancer.
Inspect the face: inspect for symmetry, features, movement and skin conditions Trigeminal neuralgia (tic douloureux), sharp, shooting, piercing facial pains. Lasts from seconds to minutes. Pain over the area of the fifth trigeminal CN (ophthalmic, maxillary, and mandibular area)
Expressions are formed by the facial muscles which are mediated by cranial nerve VII (7), any ticks or grinding of the jaw

Palpate the head: The head is normally hard and smooth, without lesion. If lesions or lumps on the head are noted, that may indicate recent trauma or a sign of cancer. Wear gloves to protect yourself from drainage or lice or whatever could be in someone's hair.

Palpate the temporal artery which should be elastic and non tender. Abnormal: hard, thick and tender with inflammation which could be seen with temporal arteries

which lay lead to blindness.

Palpate the temporomandibular joint (TMJ) to assess the TMJ place your index finger over the front of each ear as you ask the client to open the mouth. Normal- no swelling, tenderness or crepitation with movement. Mouth opens and closes fully, and lower jaw moves laterally in each direction ABNORMAL: limited range of motion, swelling, tenderness, or crepitation may indicate TMJ syndrome.

Inspection



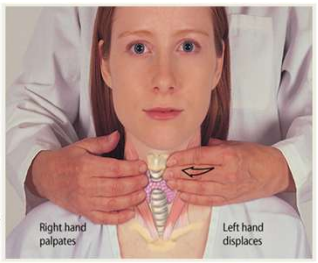
- Inspect the neck
  - Inspect the movement of the neck structures
  - Inspect the cervical vertebrae
  - Inspect range of motion

10

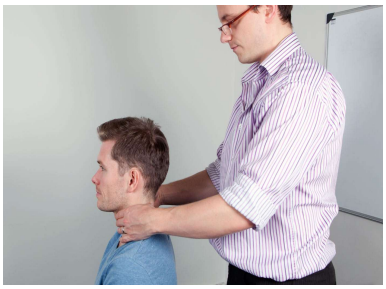
Inspect the neck: Observe the client's slightly extended neck for position, symmetry, and lumps. Any lumps or swelling is abnormal and may indicate enlarged thyroid gland, inflammation of lymph nodes or a tumor.
You also want to inspect the neck structures by asking the client to swallow a small sip of water. Observe the movement of the thyroid cartilage, thyroid gland. Any asymmetric movement or generalized enlargement of the thyroid gland is considered abnormal.
Inspect the cervical vertebrae: Ask the client to flex the neck (chin to chest) C7 Is usually visible and palpable. Swelling other than the C7 vertebrae may be abnormal.
Inspect the range of motion: Ask the client to turn the head to the right and to the left (Chin to shoulder) touch each ear to the shoulder, touch chin to chest and lift the chin to the ceiling. Should be smooth and controlled Abnormal is muscle spasms, inflammation or clerical arthritis. Stiffness, rigidity and limited mobility of the neck may be seen with advanced Ankylosing Spondylitis and meningitis

Palpation & auscultation

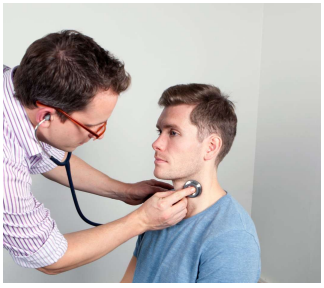
Palpate the trachea



Palpate the thyroid



Auscultate the thyroid if enlarged



11

Palpate the trachea: Place your finger in the sternal notch, feel each side of the notch and palpate the tracheal rings. The trachea is midline is normal and symmetric. Abnormal: the trachea is pushed to the side from a tumor, enlarged thyroid lobe, pneumothorax or an aortic aneurysm.
Palpate the thyroid gland: using posterior approach, stand behind the client and ask the client to lower the chin to the chest and turn the neck slightly to the right. This will relax the client's neck muscles. Then place your thumbs on the nape of the client's neck with your other fingers on wither side of the trachea below the cricoid cartilage. Use your left fingers to push the trachea to the right. The use your right fingers to feel deeply in front of the sternomastoid muscle. Normal: the thyroid gland is usually not palpable, if enlargement noted then it may be a case of hyperthyroidism, Graves' disease, or an endemic goiter, the thyroid gland may be palpated.
Auscultation: No bruits are auscultated. Abnormal: a soft blowing swishing sound auscultated over the thyroid lobes is often heard in hyperthyroidism because of an increased blood flow through the thyroid arteries.
When you auscultate a bruit which side of the stethoscope do you use? Which side is the doctor using?

Palpation of lymph nodes

- Use a gentle, circular motion with the pads of your fingers
- Palpate with both hands simultaneously to compare sides
- Lymph nodes should feel moveable, discrete, soft, & non-tender

12

Palpate the preauricular nodes, postauricular, occipital, tonsillar, submandibular and the submental nodes.
Discrete: Individually separate and distinct.
Tender, enlarged nodes suggest acute infections; usually lymph nodes are not sore or tender.

Sign & symptoms of altered thyroid

Hyperthyroidism

- Weight loss
- Fatigue
- Enlarged thyroid
- Heat intolerance
- Excessive sweating
- Nervousness
- Tachycardia
- Diarrhea
- Bulging or puffy eyes

Hypothyroidism

- Weight gain
- Fatigue
- Muscle aches
- Pale, dry skin; Thin, brittle nails
- Bradycardia
- Feeling extremely cold
- Constipation
- Irritability and depression
- Puffy face



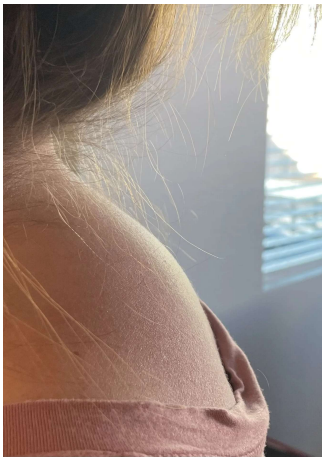
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Hyperthyroidism: Overproduction of thyroxine hormone. Increases metabolism. Common cause: Graves' disease (autoimmune issue), thyroid nodules, thyroiditis.

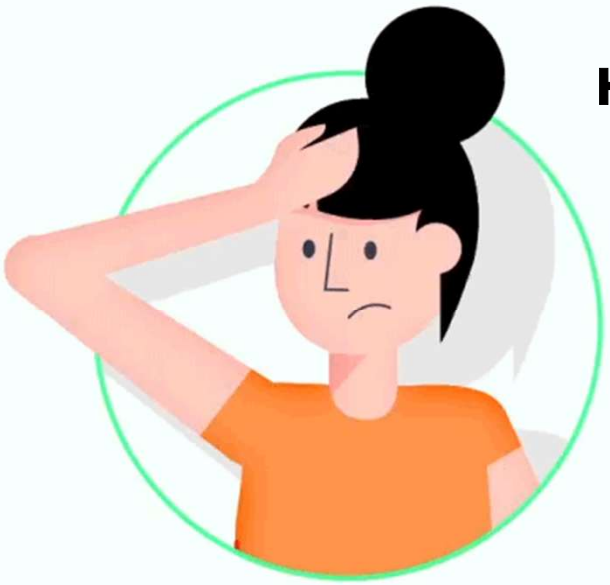
Hypothyroidism: Deficiency of thyroid hormone. Decreases metabolism. (AKA: Hashimoto's disease)

Older adult considerations

- Cervical curvature may increase from kyphosis
- Those assigned female at birth may have a fat pad over the cervical spine called a “dowager’s hump”
- Decreased ROM due to arthritis or osteoporosis
- Temporal artery pulse strength may be decreased
- Facial wrinkles are more prominent
- If palpable, thyroid may feel more nodular or irregular due to fibrotic changes
- Thyroid may be lower in the neck due to age-related structural changes



14

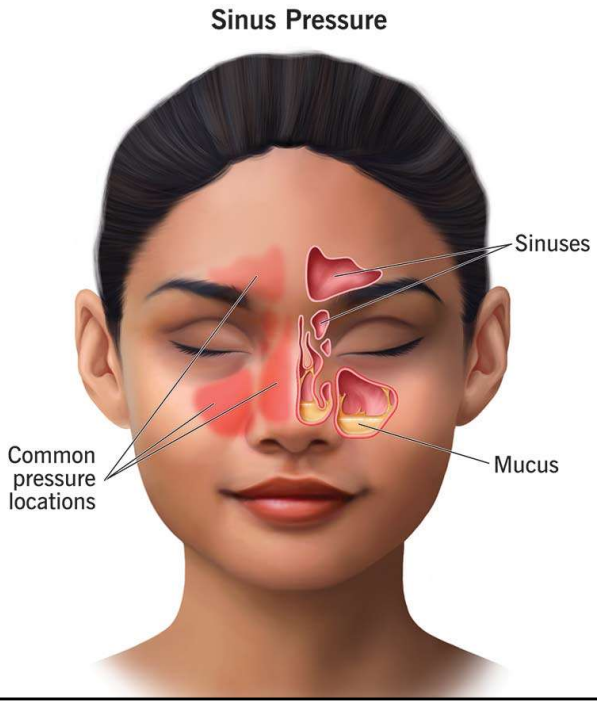


Headaches

15

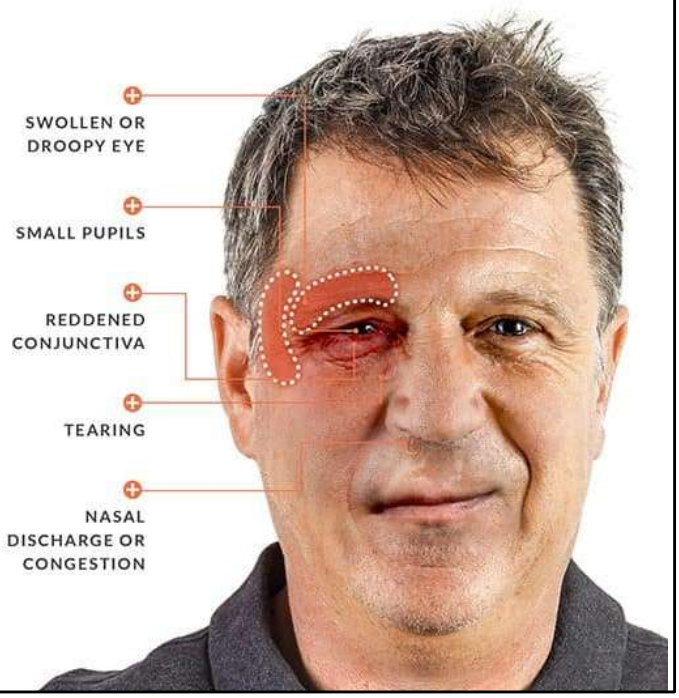
Headache: Sinus

- Deep, constant, throbbing pain
- Pressure usually across eyes or directly below
- May be tender to touch
- Tends to occur with a colds or acute sinusitis
- May occur in one area or extend along the face
- Pain tends to be worse with movement or bending forward



Headache: Cluster

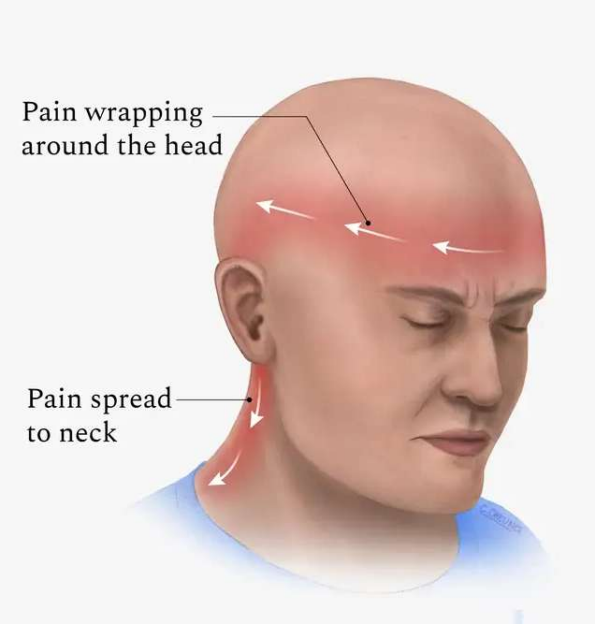
- Sudden onset of stabbing pain
- May have tearing, eyelid drooping, or runny nose
- Alcohol may precipitate headache
- Tends to occur more in those assigned male at birth
- Tends to occur at the same time of the year (seasonal)
- May last for several days then not again for months to years
- Pain is intense



17

Headache: Tension

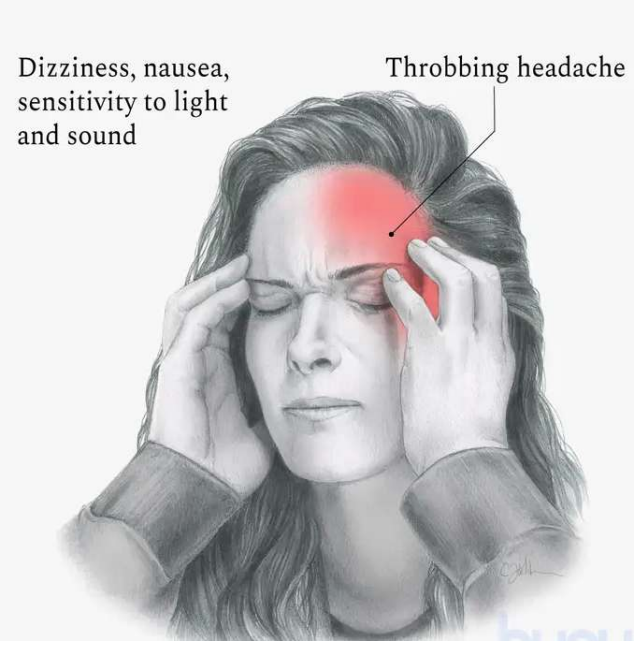
- Most common type of headache
- Tend to be dull, sense of tightness
- Occurs with lack of sleep, stress, anxiety, or depression
- Often located in the frontal, temporal, or occipital region
- Duration varies
- Aching type of pain



Headache: Migraine

- Prevents client from doing their usual daily activities
- Tend to be accompanied by nausea, vomiting, & sensitivity to light, sounds, smells
- May have a prodromal stage
- Often precipitated by stress, emotional events, anxiety, certain foods
- Tends to affect one side of the face
- May last up to 3 days
- Pain is throbbing, severe
- Thought to have genetic cause

19



Prodromal stage: Visual changes, vertigo, tinnitus, numbness or tingling of fingers/toes

Food triggers: Cheese, chocolate, alcohol

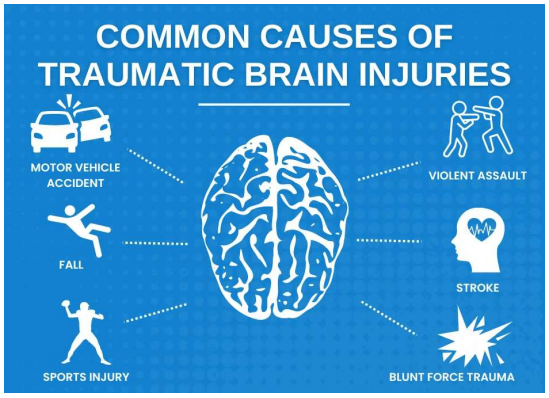
Headache: Medication overuse

- "Rebound" headache
- Dull, constant pain
- Develops in those taking large doses of pain relivers for at least 3 months (Acetaminophen, Opiates)
- Location varies
- Occurs most days
- Variable in intensity
- May also have nausea & vomiting, diarrhea, disturbed sleep, restlessness



20

Health promotion & disease prevention: TBI

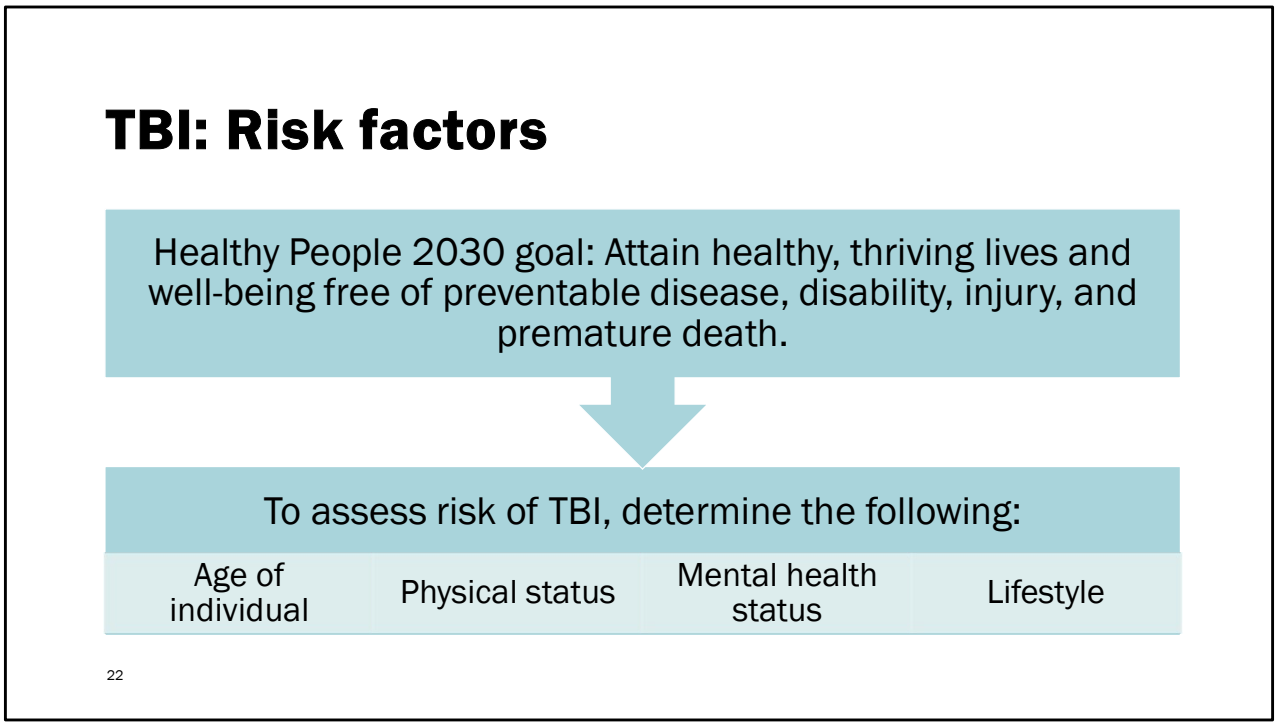


- Traumatic brain injury (TBI): From a forceful bump, jolt, or blow to the head or body, or an object that pierces the skull & enters the brain
- Short-term problems with brain function
- Severe or permanent disability or death
- Major cause of death & disability in the U.S.

21

Although not all blows to the head result in TBI, some types of TBI can cause short-term problems with brain function, affecting how the person thinks, understands, moves, communications, or acts. More severe TBI can lead to severe or permanent disability, and even death.

TBIs can be caused by falls (which lead to almost half of the TBI-related hospitalizations in the United States), firearms, motor vehicle crashes, and assaults.



OBJECTIVES

Healthy People 2030 has objectives for TBI:

Reduce fatal traumatic brain injuries
Increase the percentage of adults who resume more than half of their usual activities 5 years after TBI rehabilitation.
Reduce emergency department visits for nonfatal unintentional injuries

Risks: Falls (both infant/toddler and older adults)
Lack of parental knowledge (Can result in shaken baby syndrome)
Violence (guns, child abuse, IPV)
Alcohol
Transportation (accidents involving automobiles, motorcycles, bicycles, pedestrians)
Lack of protective gear (Helmets, seat belts)

TBI: Client education

- Protective gear
  - Seatbelts
  - Helmets
- Preventative measures to avoid falls
- Do not drive while under the influence
- Safety gates

23



Wear a seat belt when in a motor vehicle (driving or riding)
Wear a helmet to lower the chance for TBI during sports and recreation
Take preventative measures to avoid falls
Never drive while under the influence of alcohol or drugs
Children and adults should wear a helmet, or appropriate headgear

Prevent older adult falls by doing the following: Asking your doctor or pharmacist to review your medicines for any that might make you dizzy or sleepy
Having your eyes checked at least once a year (and updating your eyeglasses if needed)
Make your home safer to prevent falls (e.g., remove throw rugs, add support handles in showers and other parts of the bathroom)

Improve safety of children by doing the following:
Installing window guards to prevent young children from falling out
Using safety gates at the top and bottom of stairs for young children
Ensuring playground equipment has soft material under it (e.g., mulch, sand)

Review question

- The nurse is assessing a client's thyroid gland as part of a comprehensive physical examination. Which of the following findings should the nurse expect? (Select all that apply)

A. Palpating the thyroid in the lower half of the neck

B. Visualizing the thyroid on inspection of the neck

C. Hearing a bruit when auscultating the thyroid

D. Feeling the thyroid ascend as the client swallows

E. Finding symmetric extension off the trachea on both sides of the midline

24

Correct: the thyroid gland lies in the anterior portion of the lower half of the neck, just in front of the trachea.
Not correct: an average size thyroid gland is not visible on inspection
Not correct: A bruit indicates increased blood flow possible due to hyperthyroidism
Correct: When the client swallows a sip of water, the nurse should feel the thyroid move upward with the trachea.
Correct: The thyroid gland lies in front of the trachea and extends symmetrically to both sides of the midline.